The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.5.3 (prototype)
Log entry date	10 November 2026
Status	Released: deployed by the merge of the export-fix branch into the GitHub main branch, together with the 0.5.3 documents and this log. Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch
Previous version	0.5.2, 10 November 2026, pull requests #5 (commit 1947993) and #6 (commit 6e0de89) (see Spaghetti_Engine_Dev_Log_v0.5.2.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.5.2 (10 November 2026) to version 0.5.3 of The Spaghetti Engine. Version 0.5.3 repairs figure export across the tool and adds a figure editor, so that every figure can be exported ready for a paper.

The log documents what changed and why, how existing results are affected, how the changes were checked, and the deployment status.

2. Summary of Version 0.5.3

Area	Change
Export repaired	Exported figures no longer come out tiny or cropped, no longer depend on the size of the browser window, keep the Recharts legends, and are captured only after a chart has finished drawing.
Figure editor	A pop-up behind every Export figure button with a live preview: size in millimeters, font and text size, colors, line thickness, title and panel letter, framing and cropping, and PNG, SVG, or PDF at 300, 600, or 1200 dpi.
Paper templates	Journal, sans serif (Arial 7 pt, Okabe-Ito, PNG at 600 dpi) and Journal, serif (Times New Roman 8 pt, Tol bright, PDF at 600 dpi).
Record of 0.5.2	Version 0.5.2 was deployed by pull requests #5 (commit 1947993) and #6 (commit 6e0de89) on 10 November 2026.

3. The Changes in Detail

3.1 Export repaired

- **Tiny or cropped figures.** Figures drawn in their own coordinates were cut off or shrunk on export; every figure is now laid out again at the export width and saved whole.
- **Dependence on the window.** The size of an export followed the size of the browser window; it now follows only the chosen width and text size.
- **Legends and animation.** Recharts legends, which are not part of the chart's SVG, were lost; they are now included. A chart is captured only after its animation has finished, so no export shows a chart half drawn.

3.2 The figure editor

- **Preview and size.** The preview is the file that will be saved, with its size in millimeters and pixels and the size the text prints at. Width presets of 85, 120, 174, and 250 mm, or a custom width; a height for charts; a note above about 240 mm.

- **Style.** Font (Inter, Arial or Helvetica, Times New Roman, Georgia), text size from 5 to 14 pt, colors as on screen, Okabe-Ito, Tol bright, or greyscale, line thickness, and a white or transparent background.
- **Title, framing, and file.** Title and subtitle shown or hidden and editable, a panel letter, crop to what is drawn, a margin and trims in millimeters, an optional frame, and PNG (print size recorded), SVG (sized in millimeters, Inter embedded), or PDF (one page at the figure's size).
- **Templates and memory.** Two journal templates set every option at once; the settings are remembered between exports, and Reset to the screen style restores the defaults.
- **Figures adapted for print.** The exported prioritization matrix carries a numbered key, and the metro map uses a more compact layout.

4. Implementation Choices

Topic	Choice in the tool	Reason
Text size in print	The figure is laid out at w = W (f × 0.75) / t and scaled to W.	Labels are placed by the figure's own layout at their print size, instead of being stretched or squeezed.
Figures with a fixed aspect	Their font sizes are scaled instead.	The metro map and the transition point panels keep their proportions at every width.
Palette swap	Colors grouped by hue into families, each family mapped to the nearest unused palette color, lighter shades kept lighter.	The distinctions of the original figure survive the change of palette; greys and text are kept.
PDF output	One page at the figure's size, with the figure as a high-resolution image.	Reliable in every browser without a PDF library; SVG remains available as a vector format.

5. Effect on Existing Results

- No analysis result changes. Only exported images differ from 0.5.2.
- Version raised from 0.5.2 to 0.5.3 in package.json and package-lock.json.

6. Defects Found and Fixed

- Exports that were tiny or cropped, that depended on the browser window, that lost the Recharts legends, or that caught a chart while it was still animating (Section 3.1).

7. Verification

- **Tests:** the tests in tests/ pass.

- **Exports:** every Export figure button was used with both templates and the default settings, in PNG, SVG, and PDF, and the files were checked for size, text size, and completeness.
- **Build:** the production build completes without errors.

8. Deployment Status

- Deployed by the merge of the export-fix branch into main, with the 0.5.3 documents and this log. Netlify rebuilds the site automatically.
- **Files changed:** src/SpaghettiEngine.jsx and src/PathwayMetroMap.jsx; with the documents, package.json, package-lock.json, README.md, src/docs-manifest.json, public/sitemap.xml, and the three PDFs in public/docs, where the 0.5.3 versions replace those of 0.5.2.
- **Build note:** npm run build fails in the local OneDrive folder because of the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 to 0.5.2 logs still apply.
- A PDF export holds the figure as a high-resolution image, not as vector graphics; use SVG where a vector file is needed.
- The templates follow common journal requirements; a journal's own guidelines take precedence.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.5.3)	How to use every feature. Section 1.2 lists the changes of 0.5.3; Section 14.2 describes the figure editor and its templates.
The Spaghetti Engine - Methodological Annex (version 0.5.3)	The methods; Section 4.8 adds how figures are laid out for export and how palettes are swapped.
Spaghetti_Engine_Dev_Log_v0.5.2	The 0.5.2 release: the simplified analysis tabs and the pathways metro map.
This Development Log	What changed in 0.5.3 and the technical work behind it.

11. Next Steps

- Try the templates against the guidelines of the journals the case-study papers are submitted to.
- Describe the expert-defined sequences and the robustness families of 0.4.4 in the manual and the annex.
- Continue with the roadmap: version 0.6.0 (modelling made easier), including the Game mode on its own branch.